Tools and execution limits

GROUP QUIZ

Core questions 1-5

Choose one answer for each question. Discuss your reasons as a group before checking with the instructor. Questions 1-20 are the core set; questions 21-25 are optional. These are learning questions, not exam predictions.

1. A developer sends the lookup_case tool to the model in the request's toolConfig. What does the model learn about the tool?

A. It learns the function's source code, so it can predict the result before it asks for the tool.
B. It learns the tool's IAM role, so it can tell which tables the tool may read before it asks.
C. It learns the tool's name, its description and the shape of its input, but never the function's code.
D. It learns the caller's scope, so it can choose a case that the dispatcher will allow without refusal.

2. Each option describes a Kestrel task. Which task is better handled by ordinary application code than by a model with tools?

A. A customer gives their case reference and asks why their first bill was higher than their monthly price.
B. A customer on an open outage case needs an update that describes where things stand now.
C. Every customer whose case has waited 48 hours for their reply receives a reminder from Kestrel.
D. A first-line agent asks, in their own words, which team owns a particular open case.

3. The description of draft_reply says only "Creates a reply for a case." What is the most likely consequence?

A. The model may tell the user that the reply was sent, as nothing says that a supervisor approves it.
B. The dispatcher may refuse every draft, because it reads the description to decide which requests to allow.
C. Bedrock may refuse the request, because the API requires each description to name the tool's failure modes.
D. The model may skip the case lookup, because a short description hides the input schema from the model.

4. A developer sets the strict flag on lookup_case and proposes removing the application's own input validation. Why does the course keep it?

A. The strict flag works only for tools served over MCP, so it has no effect on Converse tool requests.
B. The application validates every tool request before execution, including requests from outside the model.
C. The strict flag doubles the cost of each tool request, so it is turned off once the course goes live.
D. Bedrock applies the schema only to the first request in a run, so later requests arrive unchecked.

5. Day 8 serves one of the assistant's tools from an MCP server. Why does the exam guide speak of standard function definitions for this kind of tool?

A. MCP requires each tool to be registered with Amazon Bedrock, which then shares one definition.
B. MCP replaces the input schema with an output schema, so the model needs only the tool's result.
C. MCP tools need no description, because the server explains each tool to the model at run time.
D. The name, description and input schema carry over almost unchanged from Converse to MCP.

Tools and execution limits

GROUP QUIZ

Core questions 6-10

6. A learner sets toolChoice to the named form for lookup_case and sends the request to a model that doesn't support it. What happens?

A. Bedrock falls back to auto, so the model decides for itself whether to ask for the tool.
B. Bedrock ignores toolChoice and runs lookup_case itself, because the tool was named.
C. The request fails with a ValidationException, so the learner should test the model first.
D. The model returns a malformed_tool_use stop reason, which the loop retries with auto.

7. A Converse response ends with the stop reason tool_use and contains a toolUse block for lookup_case. What must the application do with the block's toolUseId?

A. Copy it exactly into the matching toolResult, without parsing it, because its format can vary.
B. Parse it to find the Region that generated it, because the identifier encodes the destination.
C. Replace it with the run identifier, so the decision log and the history use one identifier.
D. Drop it from the history, because the model matches results to requests by their order.

8. A run about case_1017 reaches its third model call. What must that request contain?

A. It must contain only the latest tool result, because Bedrock stores the earlier turns of the run.
B. It must contain the question and the latest answer, because tool turns are kept by the dispatcher.
C. It must contain the tool definitions only on the first call, because Bedrock caches them for the run.
D. It must contain every message in order, with each toolUse and toolResult, and the same toolConfig.

9. A developer's lookup_case returns the text "Error: case not found" with the status success when a case is missing. Why is this a problem?

A. Bedrock rejects any toolResult whose text begins with the word Error, so the run fails.
B. The dispatcher counts the result as a repeated action, so the next lookup is refused.
C. The ledger records the error text as the answer, so the cost of the run is reported as zero.
D. The model may treat the error text as case data, because the status tells it the call worked.

10. The model asks for lookup_case with the input {"case_id": 1036}, a number, in a run scoped to case_1035. Which rule does the dispatcher's log line name?

A. It names scope, because case 1036 is outside the run's scope, which holds case_1035 only.
B. It names allowlist, because a tool request with the wrong input type isn't on the allowlist.
C. It names input_schema, because a number fails the pattern and input is checked before scope.
D. It names repeated_action, because the number matches a lookup that the run has already refused.

Tools and execution limits

GROUP QUIZ

Core questions 11-15

11. A run's deadline has passed when the model asks for issue_refund. Which rule does the dispatcher's log line name?

A. It names allowlist, because the allowlist is checked before the time left.
B. It names time_limit, because no work may start once the deadline has passed.
C. It names scope, because a refund concerns an account outside the run's scope.
D. It names approval_required, because a refund needs a supervisor's approval.

12. Another design would silently replace the model's case_id with the case that the application set. What does the course's design gain by refusing the mismatch instead?

A. It saves a model call, because the dispatcher answers from the correct case without asking again.
B. It records every attempt to reach another case as a refusal, so the attempts can be found and counted.
C. It lets the model widen the scope for the rest of the run once the customer confirms the other case.
D. It removes the need for an allowlist, because a scope check alone refuses every unknown tool name.

13. A first-line agent may read every technical support case. A run that drafts a reply on case_1026 is given the agent's whole team scope. Why is this a fault?

A. The dispatcher refuses every request in a team scope, so the run can never produce a draft.
B. A team scope disables the repeated-action check, so duplicate drafts can enter the case history.
C. The agent's role can't read case_1026, because team scopes exclude cases with customer notes.
D. The scope doesn't fit the task, so a lookup of another case passes and a draft could quote it.

14. MAX_MODEL_CALLS in config.py is 6. Why does the loop count model calls rather than tool calls?

A. Tool calls are free and instant, so only the model calls can change how long a run takes.
B. A model call is the unit that costs money, takes time and appears in the ledger as one attempt.
C. Bedrock limits each account to six model calls per run, and the loop mirrors that quota.
D. The dispatcher already counts tool calls, so a second count in the loop would conflict with it.

15. The loop checks its 180-second deadline before each model call. Which limit stops a single model call that hangs after it has started?

A. The deadline check stops it, because the loop interrupts a call as soon as the deadline passes.
B. The iteration limit stops it, because a hanging call counts as several calls while it waits.
C. The dispatcher stops it, because it refuses requests with the time_limit rule after the deadline.
D. The Day 2 client's read timeout limits how long the client waits for that one response.

Tools and execution limits

GROUP QUIZ

Core questions 16-20

16. The dispatcher refused a lookup of case_1002 with the scope rule. Later in the same run, the model sends the identical request. What does the second log line show, and where is the original reason?

A. It shows scope again, because the dispatcher checks the scope before it looks for repeats.
B. It shows allowed, because the dispatcher forgets refused requests and checks the case afresh.
C. It shows repeated_action, and the first refusal's line gives the original reason, scope.
D. It shows iteration_limit, because a refused request that returns ends the run at once.

17. Kestrel adds a guardrail with a prompt attack filter. Why must the dispatcher still check the tool requests that follow a lookup of case_1026?

A. The filter runs only on the first message of a run, so later turns reach the model unchecked.
B. The filter blocks every tool request after an attack, so the dispatcher must restore the safe ones.
C. The case note arrives in a toolResult block, and no guardrail filter evaluates tool results.
D. The filter can't read English text that contains a currency symbol, such as a £200 credit.

18. case_1035 has four notes on clean records. A run ends awaiting_approval, and a supervisor then approves the draft. How many notes does the case have after the run and after the approval?

A. It has five after the run and five after the approval, which only marks the stored draft as approved.
B. It has four after the run and five after the approval, because only approval adds the draft.
C. It has five after the run and six after the approval, which adds a second note recording the decision.
D. It has four after the run and four after the approval, because approved drafts stay in drafts.jsonl.

19. The assistant's answer gives the correct status of case_1017. How do you show from the records that the run actually looked the case up?

A. Find a lookup_case request for case_1017 and a toolResult with the same toolUseId and the status success.
B. Check that the answer quotes the case's owning team, which only a lookup could have supplied to the model.
C. Check that the run used two model calls, because every answered run with two calls made a lookup.
D. Find the answer's stop reason, end_turn, which Bedrock sets only after a successful tool result.

20. Each decision log line records input_digest, a short SHA-256 hash, instead of the request's input. Why?

A. A draft_reply input holds reply text that can quote a customer, and the hash keeps it out of the log.
B. A hash is shorter than the input, so the log stays under the size that CloudWatch accepts per line.
C. The hash lets the dispatcher rebuild the input later, if a supervisor needs to read the draft text.
D. The hash proves that the model produced the input, because constructed requests have no digest.

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GROUP QUIZ

Optional extra questions 21-25

21. The lookup tool will later run in an AWS Lambda function with its own timeout. Why does the loop still need a deadline for the whole run?

A. The Lambda timeout applies only to the first invocation in each run, not to the later ones.
B. The Lambda timeout bounds each call, up to 900 seconds, but not the total time of a run.
C. Lambda has no timeout by default, so a slow tool could run for the rest of the day.
D. A circuit breaker replaces the Lambda timeout, so the run deadline is the only limit left.

22. In a Step Functions version of the approval step, a task sends a task token to an approver and waits. What should a production design add?

A. A heartbeat timeout, HeartbeatSeconds, that fails the task when the token doesn't return in time.
B. A Choice state that approves the draft automatically when the model's confidence is above 90 per cent.
C. A second task token for the model, so that it can approve its own draft if the approver is absent.
D. A retry on the task, so that the approval request is resent to the approver every few seconds.

23. The instructor tests four wordings of a request for case_1002. Why does each wording run in a new run?

A. Each wording needs a different scope, so the application builds a new one for every request.
B. Within one run, a second request for case_1002 would meet repeated_action instead of scope.
C. The model remembers refusals across runs, so only a new run shows how the model responds.
D. The iteration limit allows one lookup per run, so a second wording would end the run at once.

24. In one run, the model declines to look up case_1002 and asks for no tool. What do the records show about this refusal?

A. The decision log has a scope line, because the dispatcher logs every case that the model mentions.
B. The ledger marks the attempt rejected, because the model refused to follow the user's request.
C. The decision log has a model_refusal line, which the loop writes when no tool is requested.
D. Only the message history shows it, because a refusal that the model chose leaves no log line.

25. Under Kestrel's goodwill credit policy, a first-line agent proposes a credit of £60.00 on a case. Who must approve it?

A. Nobody else needs to, because a first-line agent may give a goodwill credit of up to £100.00.
B. A billing manager must, because any credit above the agent's own limit goes to billing.
C. A supervisor must, because the credit is above £20.00 and within the supervisor's £150.00.
D. The assistant may approve it, if the case note shows that the customer missed a service level.